

SAMEDH SACHIN KARI

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EDUCATION

Indian Institute of Space Science & Technology, Trivandrum, India B. Tech in Electronics & Communication Engineer (Avionics) Thesis: Pose Estimation using Flash LiDAR	2021-2025 CGPA: 8.07
Namo Rims Junior College, Pune, India Higher Secondary (12th grade), Maharashtra State Board	2021 90.83 %
Millennium National School, Pune, India Secondary (10th grade), Central Board of Secondary Education (CBSE)	2019 93.80 %

PUBLICATIONS

- Samedh Sachin Kari**, A. Arockia Bazil Raj and Balasubramanian. K, *Evolutionary Developments of Today's Remote Sensing Radar Technology—Right From the Telemobiloscope: A review*, in IEEE Geoscience and Remote Sensing Magazine (IF=16.4), March 2024. [[Link](#)]
- Evolution of remote sensing radar technology from the early telemobiloscope to modern systems used in diverse applications such as military, automotive, biomedical, and Earth observation.
 - Radar system classification based on technology, frequency bands, and operating environments, highlighting different radar types like monostatic, bistatic radars.
 - Key developments in antenna technologies including monopole, dipole, phased-array, and advanced photonic and plasma antennas, and their impact on radar performance, gain, and adaptability.
 - Advances in radar waveform design, signal processing techniques, and emerging trends like AI-integrated cognitive radar and quantum radar, which enhance accuracy, resolution, and versatility of remote sensing radar systems.

RESEARCH EXPERIENCE

- Space Application Center, Indian Space Research Organisation** Jul 2025 - Present
Scientist / Engineer - 'SC', Sensor Development Area
- Assisted research and development of LiDAR systems with dynamic field of view (FOV) for autonomous docking with cooperative targets at ISRO.
 - Developed a target-adaptive LiDAR scanning algorithm that focuses only on reflective regions, improving scanning efficiency.
 - Supposed to reduce scan time by 97% and enhanced data acquisition performance through algorithmic optimization for real-time LiDAR operations.
- U. R. Rao Satellite Center, Indian Space Research Organisation** Jan 2025 - Apr 2025
Project Intern, Control Dynamics Design Group [[Link](#)]
- Developed a pose estimation algorithm for target-chaser systems using Flash LiDAR data, determining 6-DOF position and orientation from point cloud inputs.
 - Implemented methods to estimate linear and angular velocities from successive LiDAR frames, enhancing real-time tracking accuracy.
 - Applied Kalman filtering to refine pose and velocity estimates, improving stability and noise resilience of the system, and achieved accuracy of approx. 2 mm for position and 0.6° for orientation.
 - Integrated the final pose and motion estimates as control inputs within a target-chaser simulation, implementing the full pipeline in C++.
- U. R. Rao Satellite Center, Indian Space Research Organisation** May 2024 - Jul 2024
Intern, Control Dynamics Design Group [[Link](#)]

- Worked on ‘Mathematical Modelling and Preliminary Data Processing of LiDAR’.
- Developed a comprehensive mathematical model for every component of Flash LiDAR system.
- Implemented the model as a Finite-Difference Time-Domain (FDTD) simulation in C for analysis.
- Created an original LiDAR data clustering and preprocessing algorithm that discards about 65% noisy data early, accelerating subsequent computations.

Defence Institute of Advanced Technology, DRDO

May 2023 - July 2023

Research Intern, Radar Systems Laboratory

- Conducted study on ‘Evolution of Remote Sensing and Radar Technology’, focusing on advancements in Photonics Radar systems.
- Authored and published a review paper on developments in radar and remote sensing technologies.
- Assisted research and gained foundational knowledge in emerging photonic radar systems.

SCHOLASTIC ACHIEVEMENTS

- All India Rank **452** in GATE 2024 (ECE) out of 63,092 candidates (**99.28** percentile) [\[Link\]](#)
- **99.4** percentile overall with **99.99** subject percentile for mathematics in JEE 2021
- Regional Mathematics Olympiad (RMO) Awardee 2018 (Top 30 in Maharashtra region and top 300 among 10,000+ candidates across India)
- Gold Honour in International Youth Math Challenge (**Top 1 %** international participants)

KEY LAB PROJECTS

Vehicle Overspeeding Sensor

2022

Analog Electronic Circuits and E-CAD Lab

[\[Link\]](#)

- Designed an analog vehicle overspeeding detection system using basic sensors and discrete circuits to monitor whether vehicles exceeded a preset speed limit.
- Implemented with two lasers and photodiodes placed at a fixed distance, where beam interruptions by the car triggered multivibrator circuits calibrated to detect speeds relative to a threshold.
- Developed a functional prototype capable of accurately indicating whether a vehicle was traveling above or below 80 km/h, demonstrating reliable real-time speed assessment.

Digital Clock using FPGA

2023

Digital Electronics and VLSI Lab

[\[Link\]](#)

- Designed a digital clock in Verilog, including time-keeping logic, counters, and display modules.
- Implemented and tested the design on an FPGA and 7-segment display.

Smart Irrigation System

2023

Computer Architecture

[\[Link\]](#)

- Developed a smart irrigation system using Verilog to monitor soil moisture, temperature, and atmospheric humidity, enabling automated water management for agricultural applications.
- Interfaced soil-moisture, temperature, and humidity sensors through ADCs and programmed an FPGA to process sensor data and display system status using 7-segment indicators.
- Demonstrated a working prototype capable of interpreting multi-sensor environmental inputs and automating irrigation decisions, showcasing scalable logic for water-supply control.

Smart Wallet

2024

Computer Networks Lab

[\[Link\]](#)

- Designed a Wi-Fi-enabled smart wallet system to interface with a mobile application, enabling real-time monitoring of wallet usage and card presence.

- Integrated piezoelectric sensors to detect card presence and used a microprocessor with a Wi-Fi module to transmit sensor data to a mobile device, where usage information was logged.
- Delivered a fully functioning prototype demonstrating reliable sensing, data transmission, and mobile logging, showing clear potential for miniaturization into a practical smart wallet.

RADAR based Position Estimation of co-orbiting Satellites

2024

Navigation Systems and Sensors Lab

[\[Link\]](#)

- Developed a MATLAB-based proof of concept to determine the position of a co-orbiting satellite using radar range measurements and orbital prediction.
- Modeled a radar-equipped satellite capable of estimating distance to a neighboring satellite and predicting its own orbit from position-velocity data; computed the co-orbiting satellite's location from the intersection of orbital paths and radar-derived ranges.
- Demonstrated a functional simulation in MATLAB validating the feasibility of radar-assisted relative satellite positioning, establishing groundwork for more advanced space-situational awareness systems.

COURSE WORK

- **Signal Processing and Control Systems:** Signals and Systems, Control Systems, Digital Signal Processing, Machine Learning, Digital Image Processing, Radar Signal Processing, Navigation Systems
- **Communication Systems:** Analog Communication Systems, Digital Communication Systems, RF and Microwave Communication, Satellite and Optical Communication
- **Electronics:** Semiconductor Devices, Network Analysis, Analog Electronic Circuits, Digital Electronics, VLSI Design and Technology, Power Electronics, Instrumentation, RF Integrated Circuits
- **Mathematics:** Linear Algebra, Complex Analysis, Calculus, Ordinary and Partial Differential Equations, Integral Transforms, Probability and Statistics, Numerical Methods

CERTIFICATIONS

- Data Structures in C++ by Coding Ninjas : Online Certification [\[Link\]](#)
- Python : Python@5x Certification by AI-ML Club - IIST

TECHNICAL SKILLS

Programming Languages	C, C++, Python
Hardware Languages	Verilog
Tools	Linux, L ^A T _E X, Microsoft Office

HOBBIES AND EXTRACURRICULAR

Music

- Level 4 Certification in Indian Classical Music by Gandharva Mahavidyalaya.
- Proficient in playing Harmonium and Flute. Supporting research on 10-finger Dhrupad Flute.
- Resources and Inventory Manager at Neptunes Music Club in IIST.
- Organised, participated and won intercollege competitions while representing the band 'Aaroh'.

Sanskrit

- Completed 10 + 2 Education with Sanskrit as second language.
- An avid reader and enthusiast of the Sanskrit language, which I plan to pursue further.

Finance

- Chief Treasurer and Accountant for Dhanak 2024, Cultural Fest of IIST.
- Founding Member at the Money Minds Club, Finance Club of IIST.